

Panoramica del corso

Astrofisica Multi-Messaggio: dalle Onde Gravitazionali ai Raggi Gamma

42 ore divise in

- 14 ore – *Rivelatori interferometrici di onde gravitazionali* prof. Mateusz Bawaj
- 14 ore – *Radiazione elettromagnetica e Neutrini* prof.ssa Sara Cutini
- 14 ore – *Astronomia multimessaggera* prof. Giuseppe Greco



Astrofisica Multi-Messaggio:
dalle Onde Gravitazionali
ai Raggi Gamma

Onde Gravitazionali

Introduzione

corso opzionale per la laurea magistrale in fisica [LM-17] FISICA

6CFU – 42 ore

Mateusz Bawaj mateusz.bawaj@unipg.it



Libri di testo

- Peter Saulson, *Fundamentals of interferometric gravitational wave detectors*, World Scientific, 2017
- Clifford M. Will, *Theory and experiment in gravitational physics*, Cambridge University Press, 1993

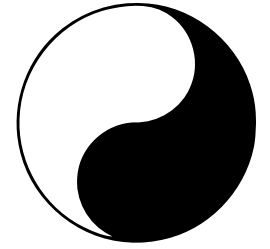
Materiali integrativi

- Sito Unistudium: <https://www.unistudium.unipg.it/>

Piano del modulo

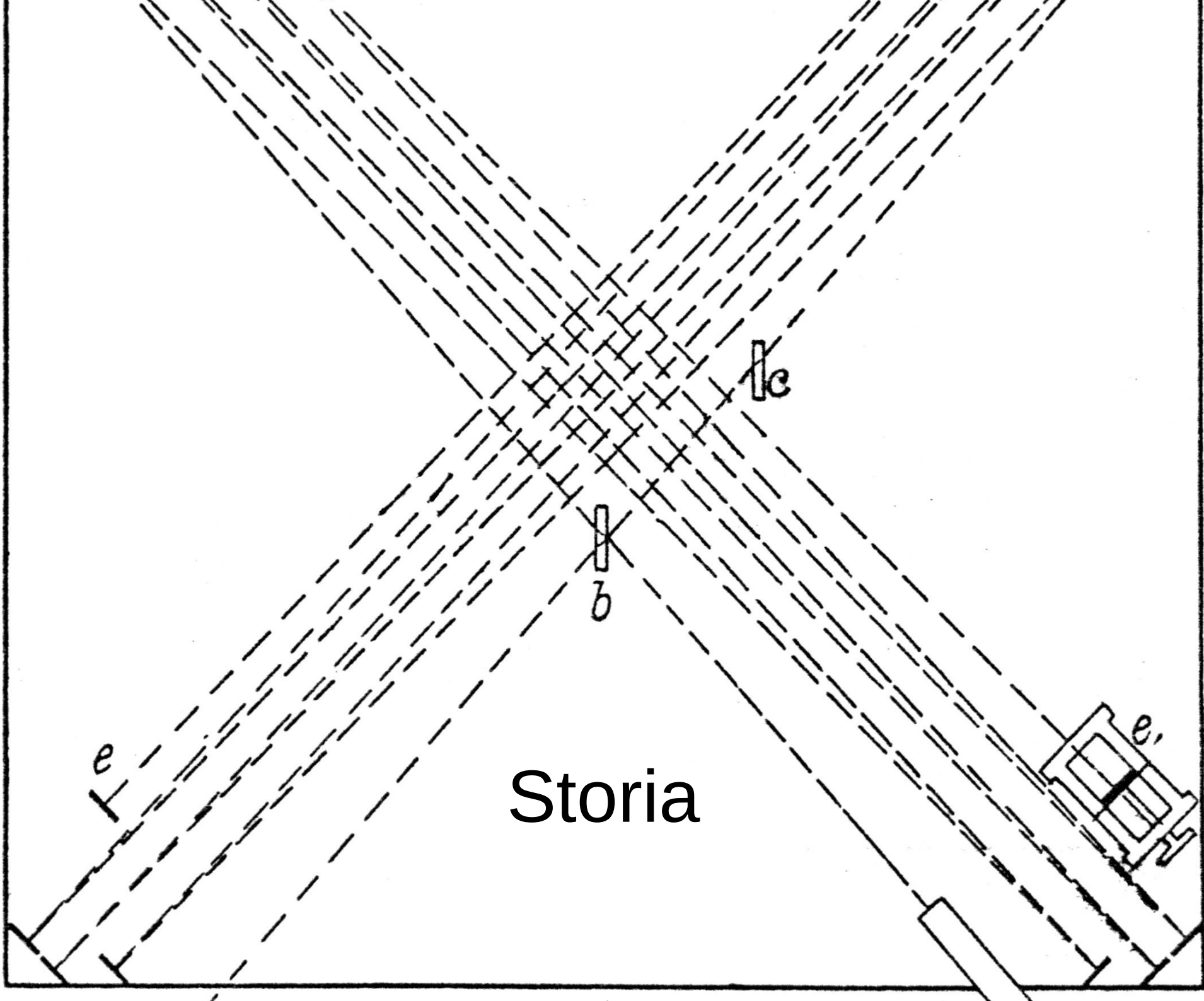
- 1) introduzione
- 2) storia
- 3) modo per misurare onde gravitazionali
- 4) rivelatori esistenti
- 5) analisi di segnali gravitazionali

Theory & Experiment

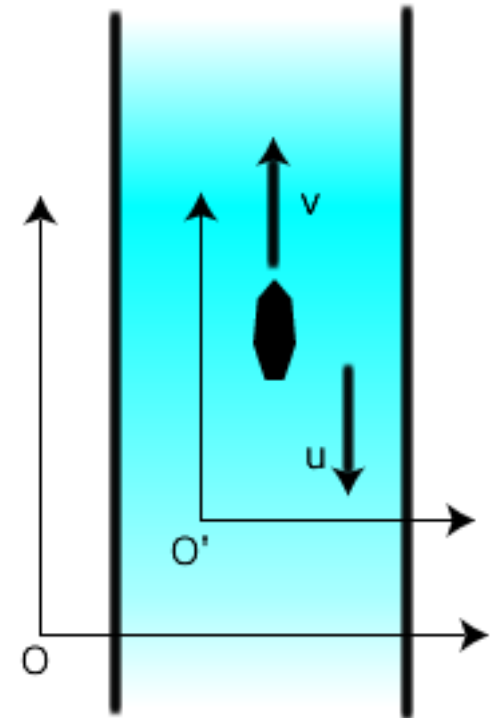
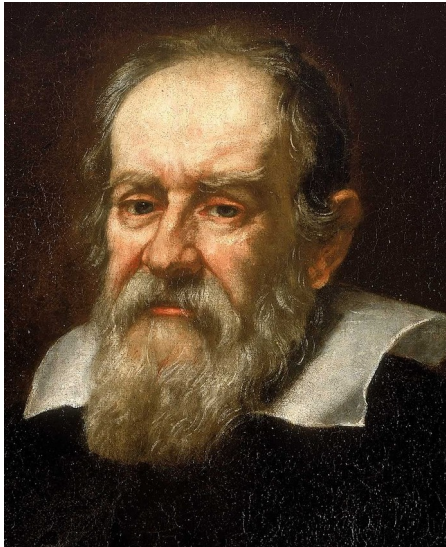


"The difference between a good experiment and a good theory is in the fact that the theory gets old quickly and it is replaced by another one, based on more perfect ideas. It will be forgotten quickly. The experiment is something else. The experiment, which has been thought well and performed carefully, will step in the science forever. It will become its part. It is possible to explain such experiment differently in different periods of times." P. L. Kapica

Storia



Relatività



leggi della meccanica hanno la stessa forma nei sistemi di riferimento inerziali



Relatività veloce

- esiste una velocità massima osservabile in ogni sistema di riferimento
- leggi della fisica sono le stesse in tutti sistemi inerziali di riferimento



Albert Einstein come l'impiegato dell'Ufficio Brevetti Svizzero (Berna, 1905)



Onde Gravitazionali - Introduzione

Relatività ristretta

ancora per un gioco

Materiali integrativi: effetto Doppler, dilatazione del tempo, contrazione delle lunghezze



A slower speed of light

Credits to MIT Game Lab

Relatività generale

- pubblicata per includere la gravitazione nella nuova relatività
- gravitazione non è più una forza ma la caratteristica di spazio-tempo

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

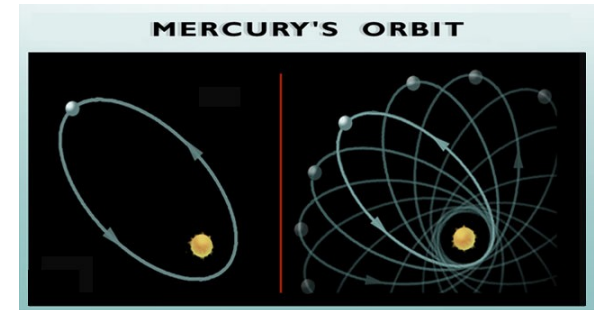
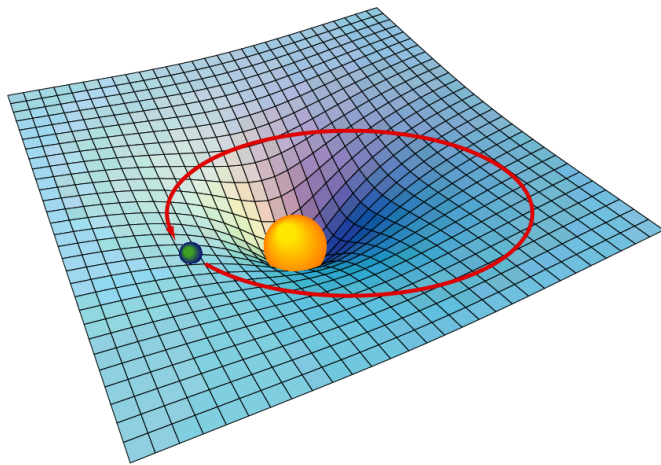
Equazione di campo di Einstein



Relatività generale

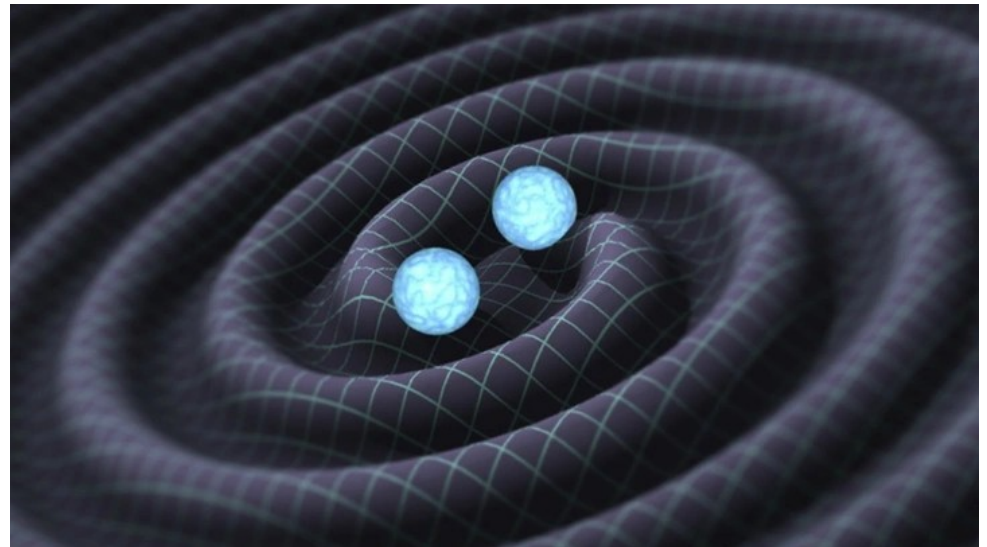
le conseguenze

- lo spazio-tempo non è uniforme ma può essere curvato, tutti gli oggetti percepiscono questa curvatura



Sospetto delle Onde gravitazionali

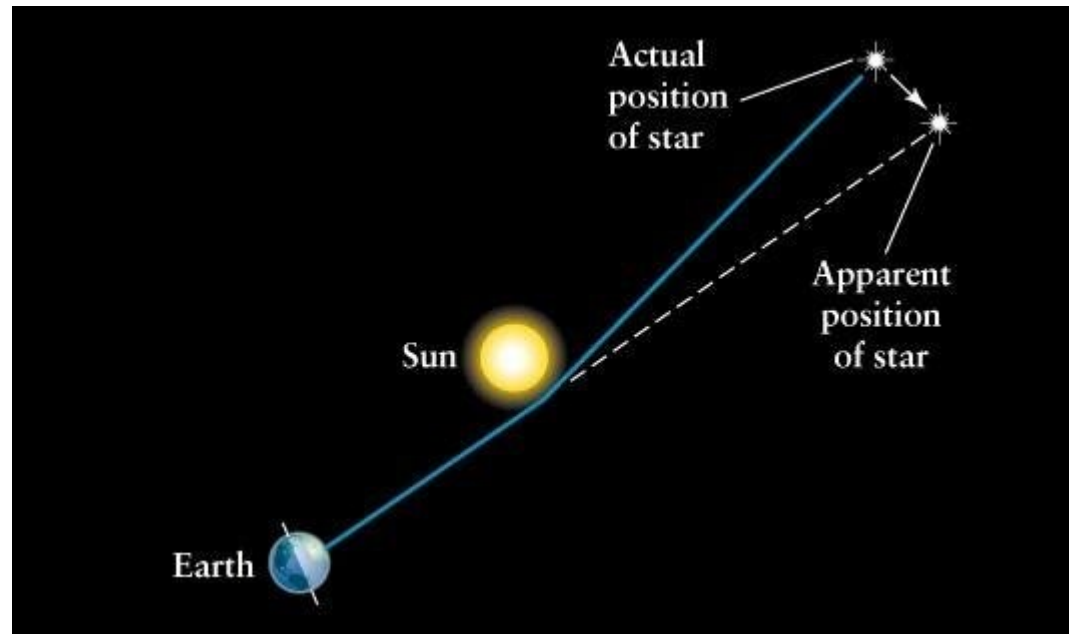
- potrebbero esistere le onde gravitazionali generate dalle masse che accelerano bruscamente
- le onde gravitazionali viaggierebbero alla velocità della luce



Relatività generale

la conferma

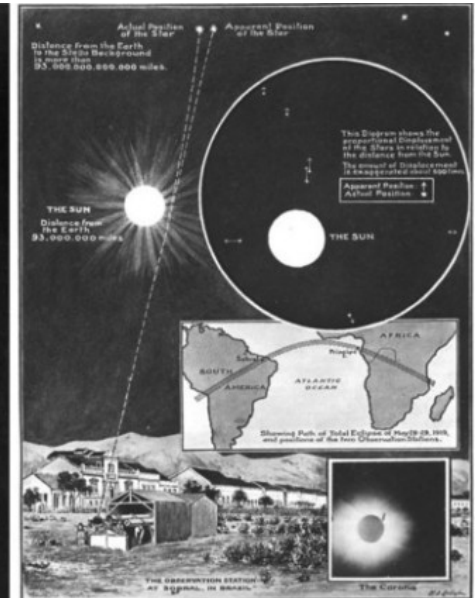
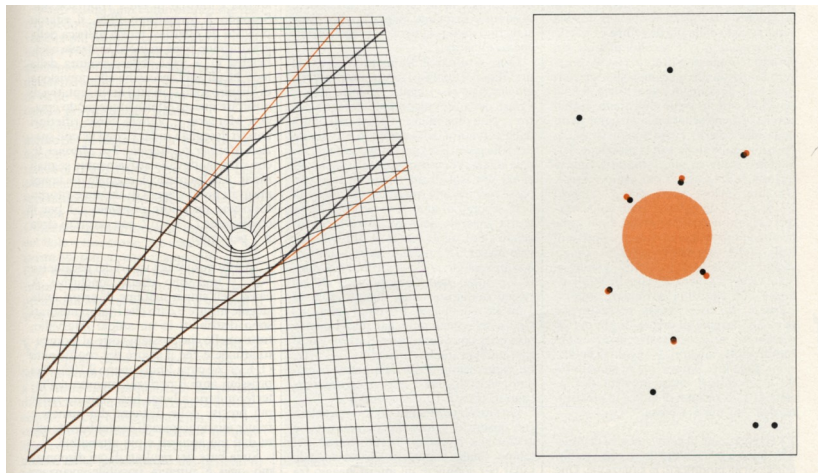
Sir Arthur Eddington & Solar Eclipse Experiment



Relatività generale

la conferma

Sir Arthur Eddington & Solar Eclipse Experiment



1632
GALILEO

ISAAC NEWTON
1686

1905
RELATIVITÀ
RISTRETTA

RELATIVITÀ
GENERALE
1915

1919
CONFERMA DI
EDDINGTON

CONFERENZA
CHAPEL HILL
1957

1960
BARRE
RISONANTI

Relativity in GPS

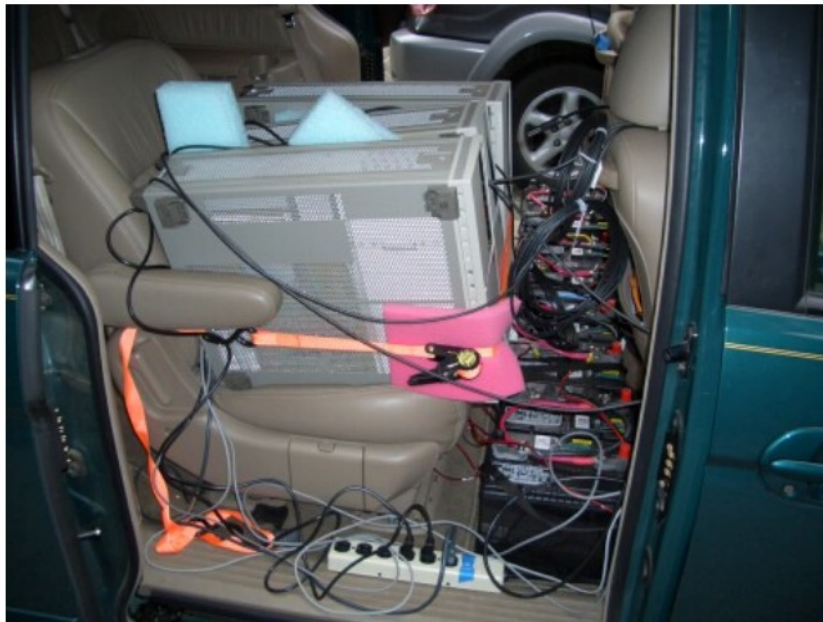
- GPS orbits are extreme:
 - speed 14,000 kph, 8,700 mph (Mach 12)
 - altitude 20,000 km, 12,500 miles ($\sim 3 \times R_e$)
 - N.B. GPS speed is only 0.000013 c, “13 μ c”
- Large relativistic effects occur:
 - $\Delta f/f$ is -8.4×10^{-11} , or $-7.3 \mu\text{s/day}$ (kinematic)
 - $\Delta f/f$ is $+5.3 \times 10^{-10}$, or $+45.6 \mu\text{s/day}$ (gravitational)
 - $\Delta f/f$ is $+4.4 \times 10^{-10}$, or $+38.3 \mu\text{s/day}$ (net effect)
 - 1.023 MHz set to 10.22999999954326 MHz

Effetto Sagnac



The GREAT trip, day 1

- 3 clocks in the middle. Batteries on the floor. Monitoring instruments in front.



November 2018



SCPNT/Stanford

53

Time dilation: *prediction*

- Home clock and mountain clock elevations
- 5400 ft – 1000 ft = 4400 ft (1340 m)
- $\Delta f/f = 1.46 \times 10^{-13}$ times 42 hours = +22 ns



November 2018



SCPNT/Stanford

62

Einstein celebrità

REVOLUTION IN SCIENCE.

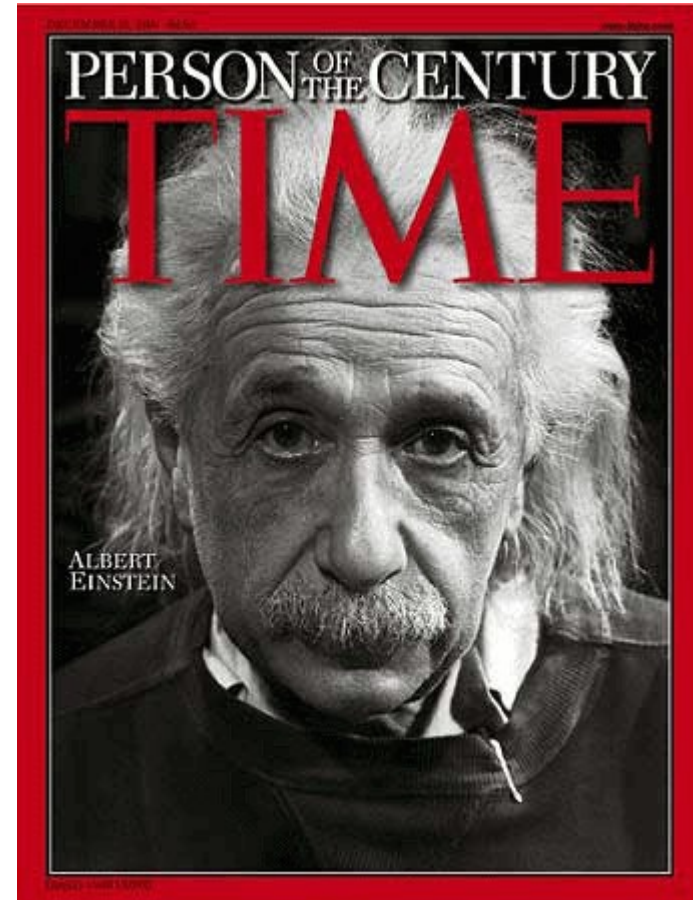
NEW THEORY OF THE UNIVERSE.

NEWTONIAN IDEAS OVERTHROWN.

Yesterday afternoon in the rooms of the Royal Society, at a joint session of the Royal and Astronomical Societies, the results obtained by British observers of the total solar eclipse of May 29 were discussed.

The greatest possible interest had been aroused in scientific circles by the hope that rival theories of a fundamental physical problem would be put to the test, and there was a very large attendance of astronomers and physicists. It was generally accepted that the observations were decisive in the verifying of the prediction of the famous physicist, Einstein, stated by the President of the Royal Society as being the most remarkable scientific event since the discovery of the predicted existence of the planet Neptune. But there was differ.

The Times, 7 novembre 1919



La storia dell'esperimento o della teoria

“Conference on the Role of Gravitation in Physic” conosciuta come Chapel Hill Conference (gennaio 1957)

organizzata da U.S. Air Force

ha raccolto gli esperti della fisica per discutere la gravitazione



Felix Pirani

- fisico teorico che con il suo mentore Hermann Bondi ha presentato durante la conferenza Chaper Hill la soluzione delle equazioni di Einstein
- il suo approccio basava sulla rivelazione non sulla generazione dell'onda



1928 – 2015

CONFERENZA
CHAPEL HILL
1957

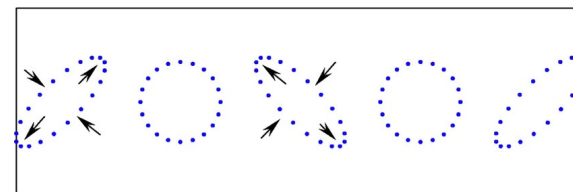
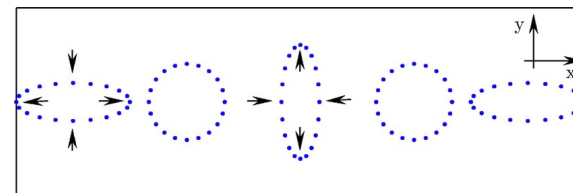
1632
GALILEO

ISAAC NEWTON
1686

1905
RELATIVITÀ
RISTRETTA

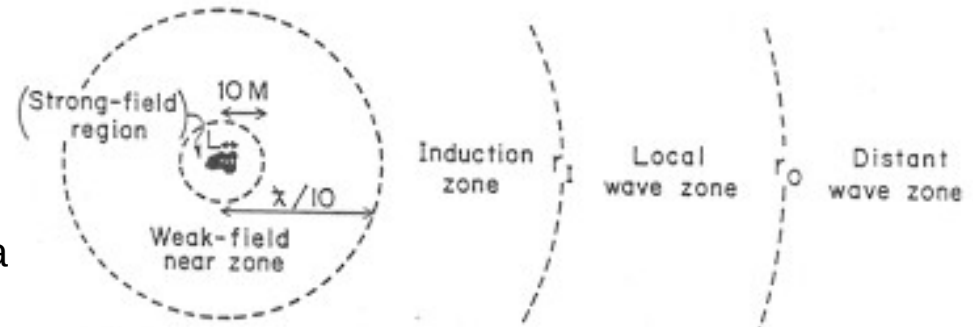
1919
CONFERMA DI
EDDINGTON

1960
BARRE
RISONANTI



Soluzioni per astrofisica

Risolvendo le equazioni di campo in base alla distanza dalla sorgente del campo distinguiamo quattro zone:



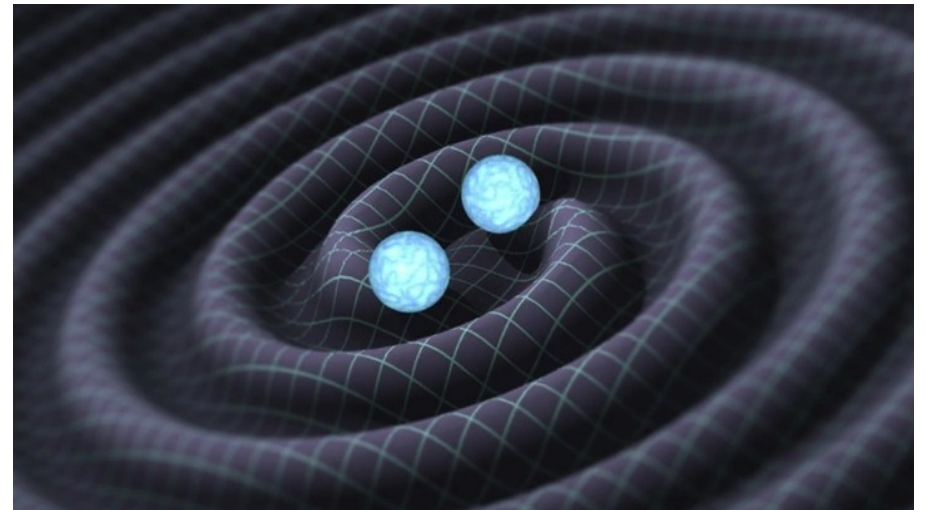
- 1) la zona di campo gravitazionale forte: in questa zona la gravita è ovviamente forte. Le soluzioni analitiche non esistono e ci rivolgiamo alla modellazione numerica in pieno regime non lineare.
 - oltre la distanza $r \sim 10L$ comincia la zona di campo debole dove possiamo considerare la gravita come debole.
 - da questa distanza, il problema può essere analizzato usando le soluzioni linearizzate nello spazio-tempo piatto.
- 2) la zona di campo debole vicina: $10L \leq r \ll \lambda/2 \pi$. Qui il contributo alla curvatura proveniente dalla sorgente è ancora non trascurabile Il campo gravitazionale può essere studiato usando espansione post-newtoniana delle equazioni di Einstein.
- 3) la zona di onda locale: da questa distanza in poi è ragionevole parlare delle onde gravitazionali. La trasformazione in onda avviene intorno alla $r \sim \lambda/2 \pi$. Altre sorgenti di gravita sono distanti e quindi non rilevanti per le analisi. Onde gravitazionali si propagano liberamente
- 4) la zona di onda distante: $r \gg \lambda/2 \pi$. Questa zona copre l'universo lontano. Ottica geometrica è uno strumento adatto per descrivere la propagazione dell'onda. I sorgenti che incontra l'onda possono essere rilevanti. Bisogna considerare gli effetti di lente gravitazionale.



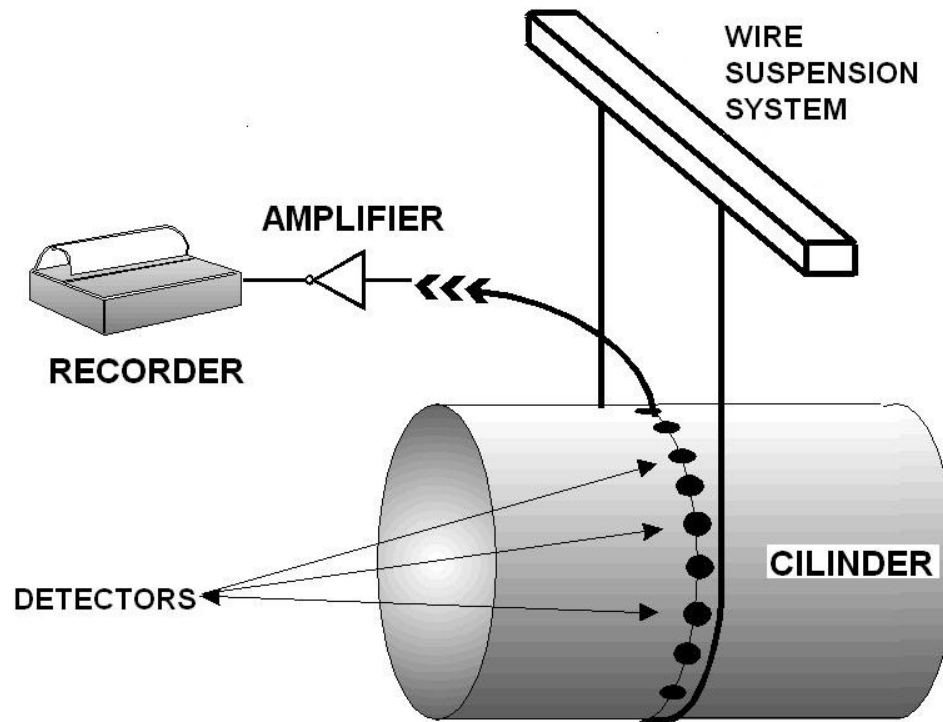
Onde gravitazionali

sorgenti

- le stelle binari sono un buon esempio
- il sistema binario fatto da due stelle di neutroni è ancora migliore
- il sistema fatto di due buchi neri è il migliore perché contiene più massa e accelera più velocemente
- le onde gravitazionali, al contrario della radiazione elettromagnetica, non vengono assorbite che permette a osservare il cosmo meglio che attraverso dei telescopi



I primi esperimenti

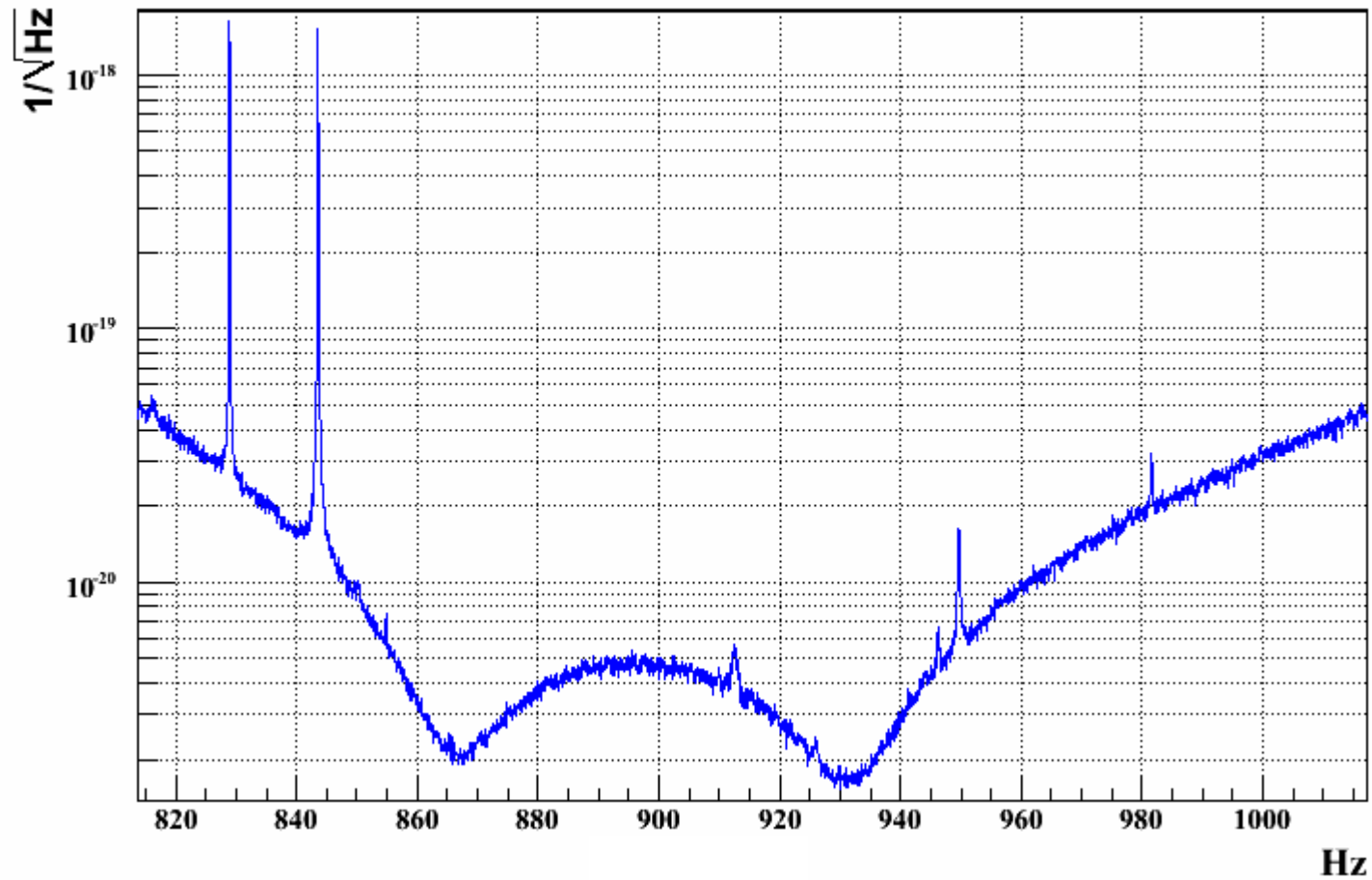


Joseph Weber lavorando con la barra



AURIGA

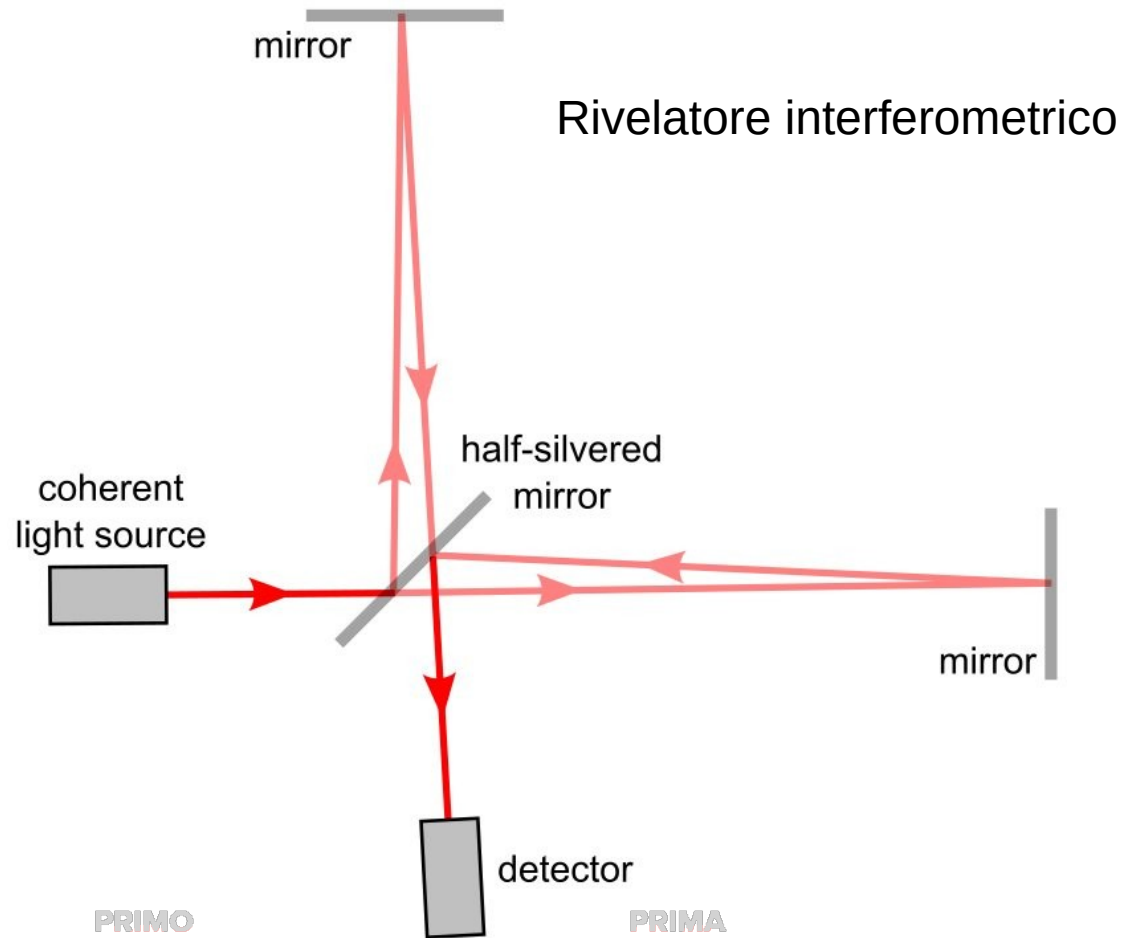
Status of the AURIGA detector as May 2009



L'idea migliore

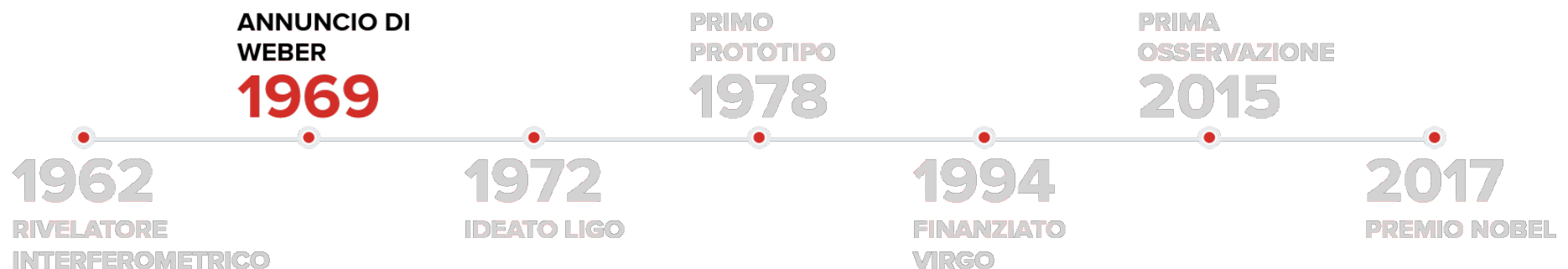


V. I. Pustovoi



Il primo annuncio di osservazione

- Nel 1969, Weber annuncia la prima osservazione dell'onda gravitazionale basandosi su una rivelazione simultanea da due rivelatori
- Il segnale misurato sarebbe sorprendentemente forte
- L'annuncio attrae molti altri sperimentatori a indagare sul campo delle barre di Weber come: gruppo di Glasgow, Roma, Frascati, Monaco di Baviera, Bell Labs o anche IBM

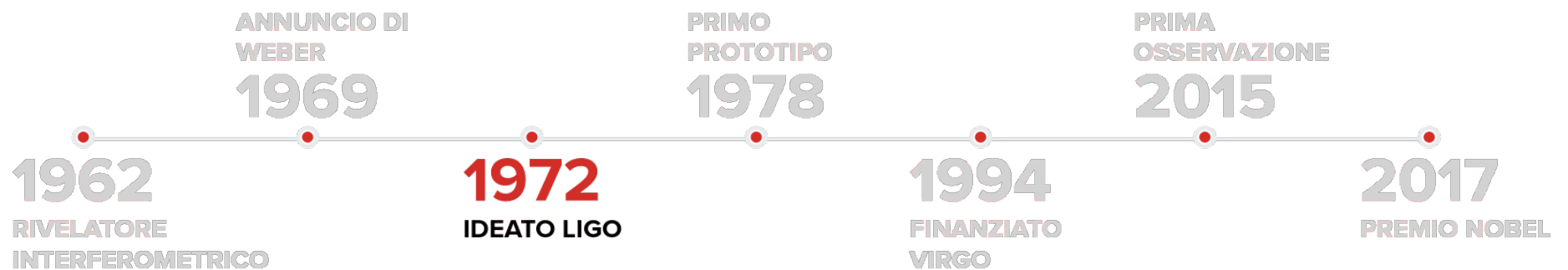


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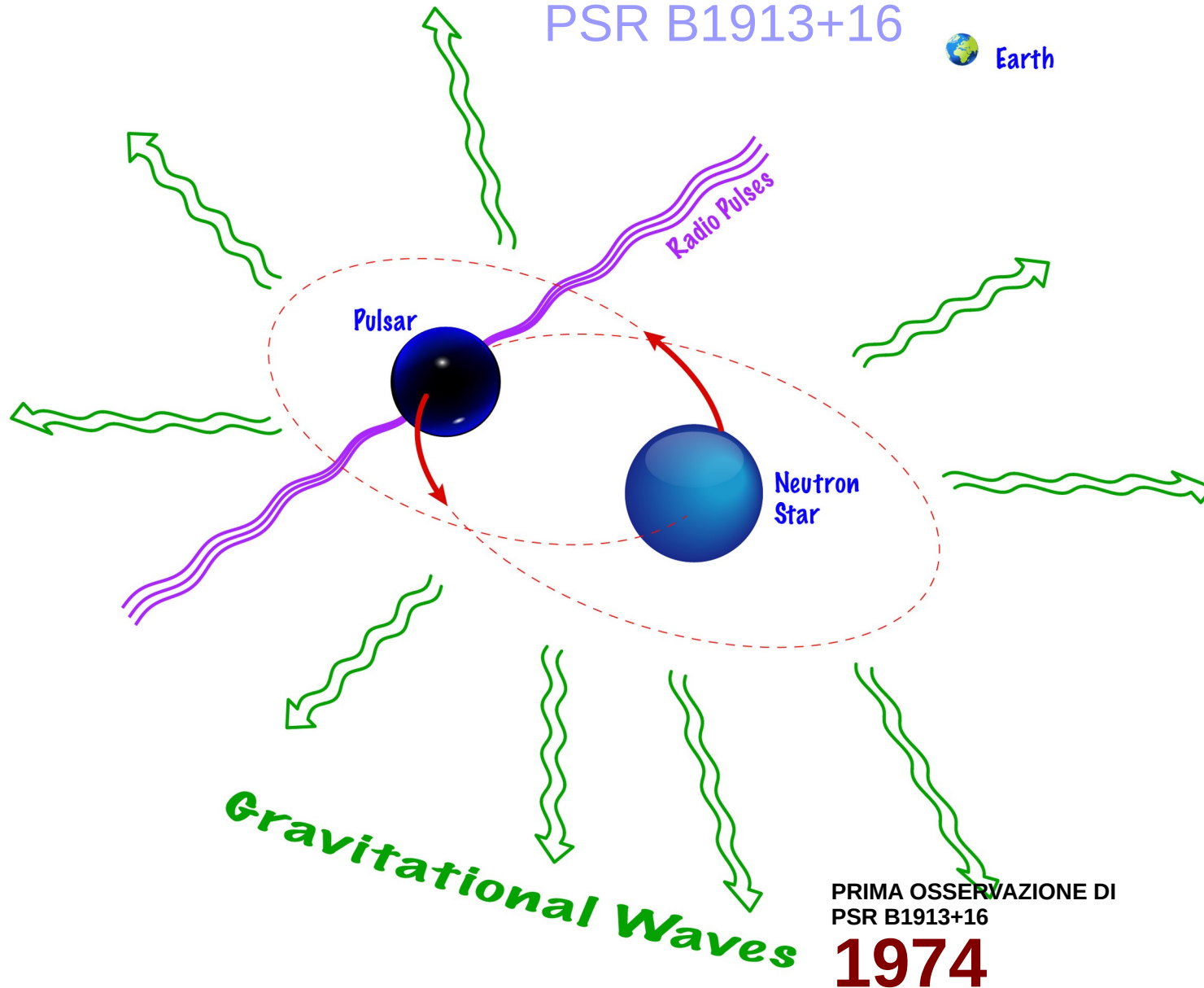


Fondazione di LIGO



Hulse–Taylor binary pulsar

PSR B1913+16



I primi prototipi

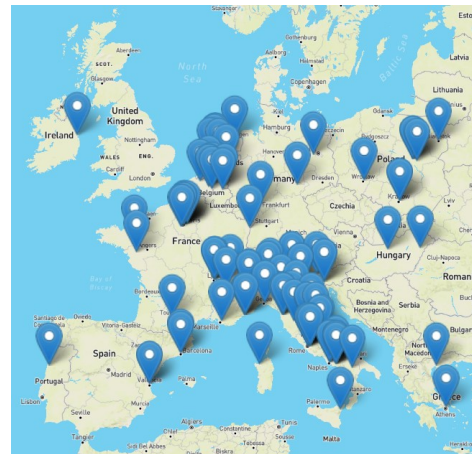
- Laboratorio Hughes Research Lab (nel 1978)
- Max Planck Institute for Astrophysics a Monaco (prototipo di 3m nel 1975)



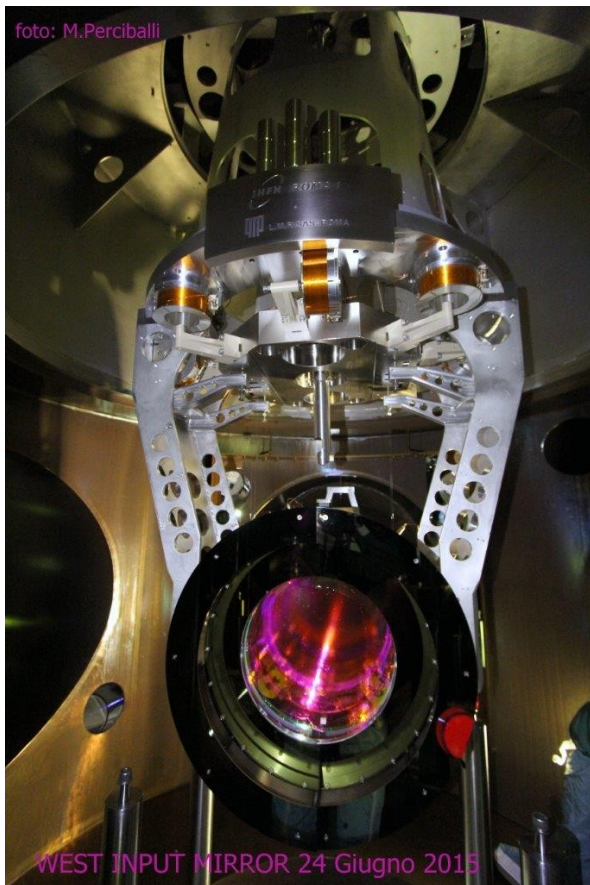
Costruzione di Virgo e LIGO

- progetto Virgo approvato dalla Centre National de la Recherche Scientifique (CNRS) e Istituto Nazionale di Fisica Nucleare (INFN)

AL 10/2020

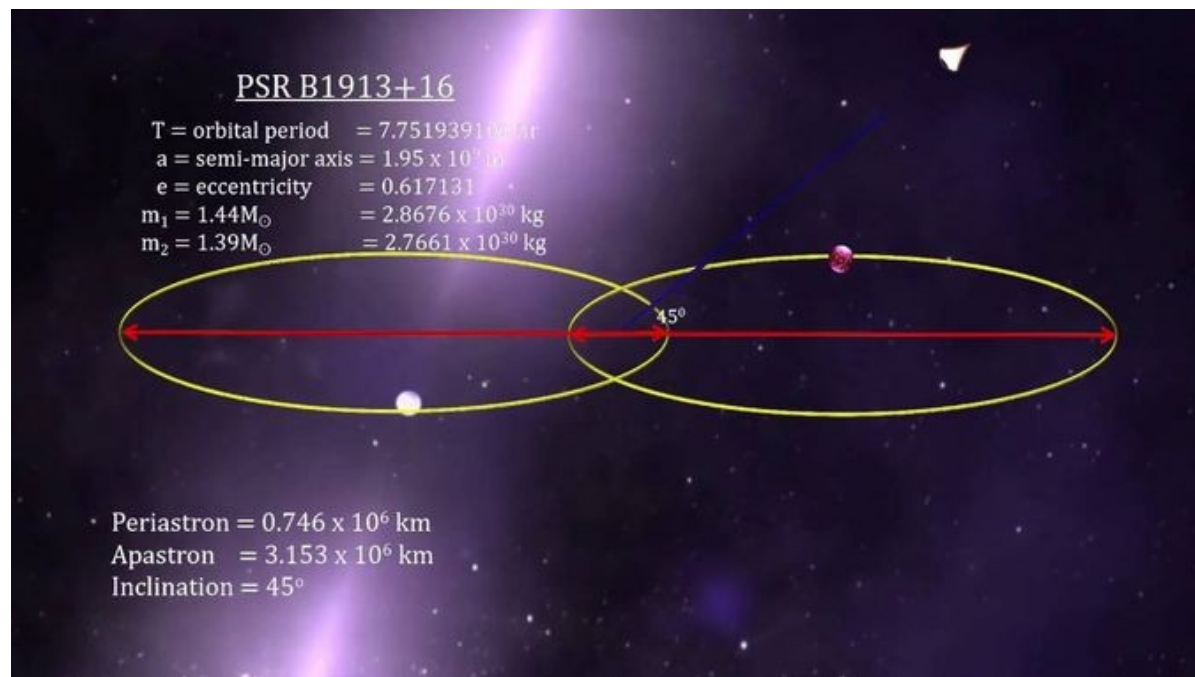
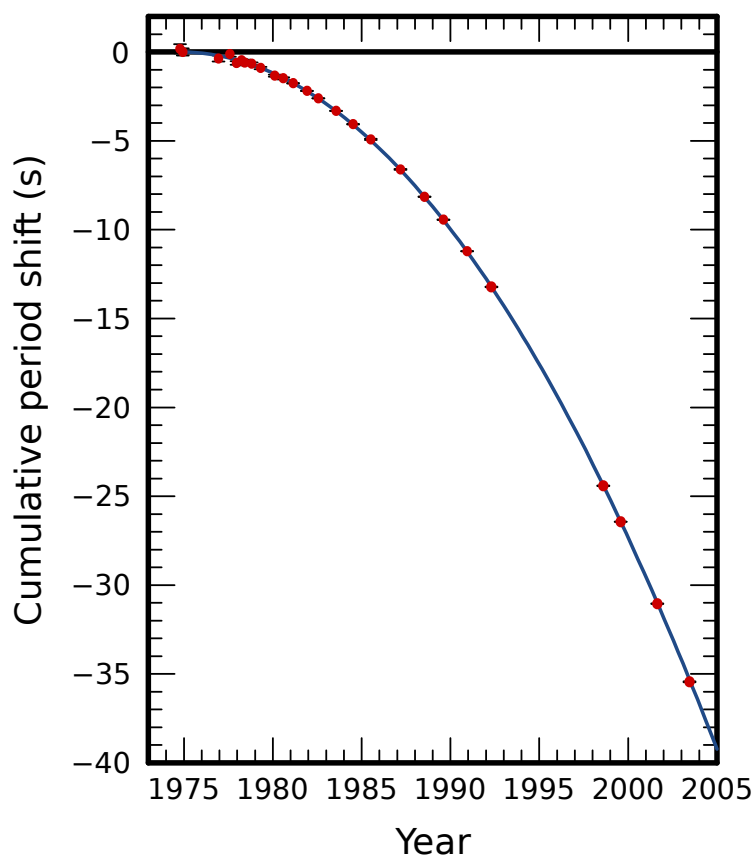


Costruzione di Virgo e LIGO



Analisi dei dati del pulsar di Hulse–Taylor

Andamento dell'orbita dalla data della scoperta fino al 2004 coincidono con il modello della perdita di energia basato sulla teoria della relatività generale

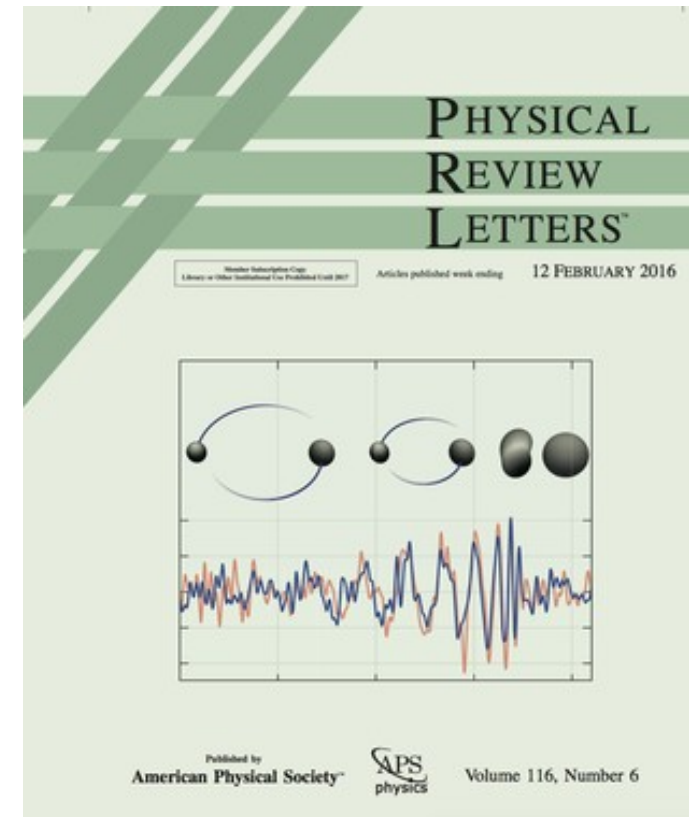
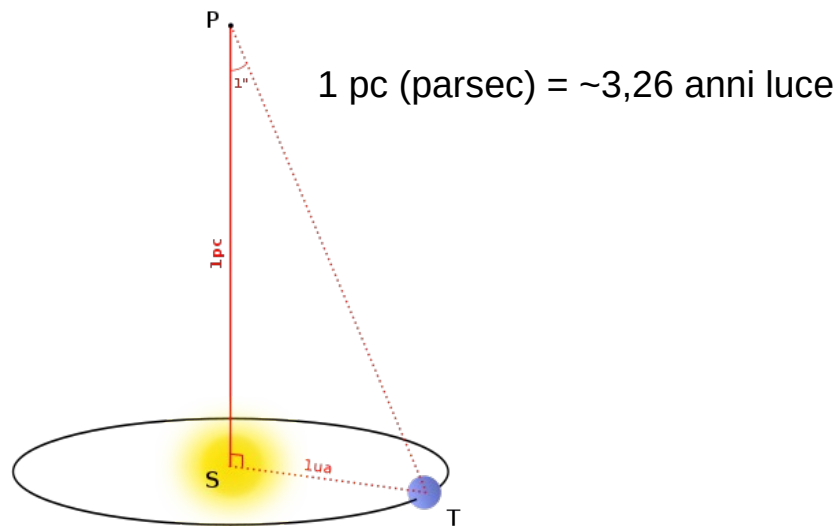


PRIMA MISURA CONFORME
CON EMISSIONE DI ONDE GW

2004

La prima osservazione GW150914

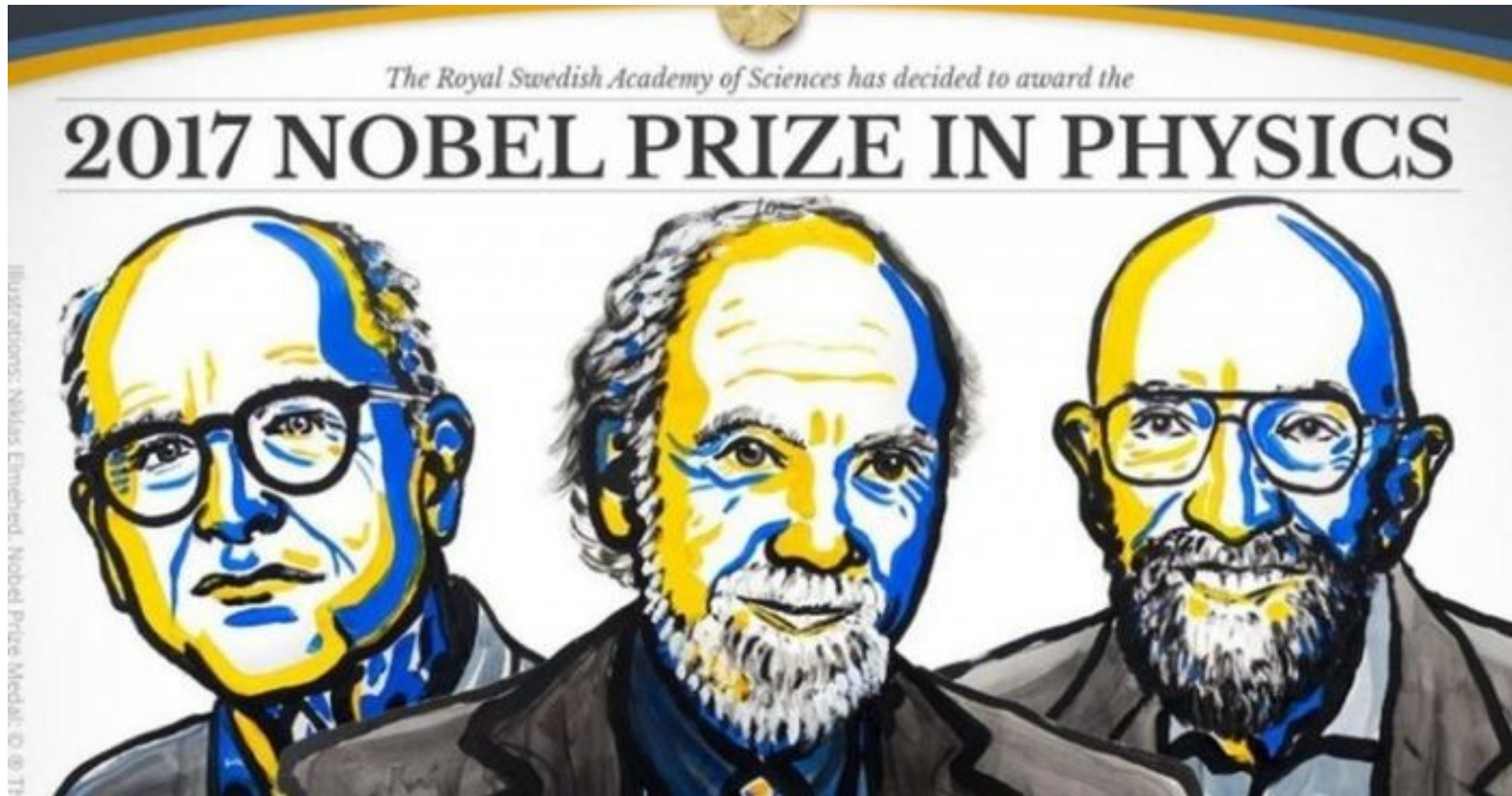
Coalescenza di due buchi neri
 $36 M_{\odot} + 29 M_{\odot}$ alla distanza di 460 Mpc



How Scientists Reacted to Gravitational Wave Detection? (video)



Premio Nobel



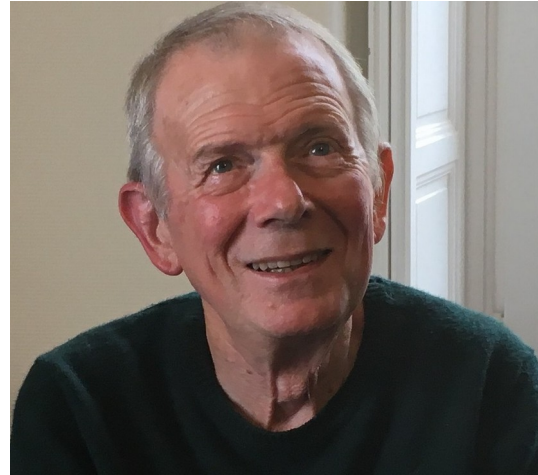
Rainer Weiss, Barry C. Barish, Kip S. Thorne



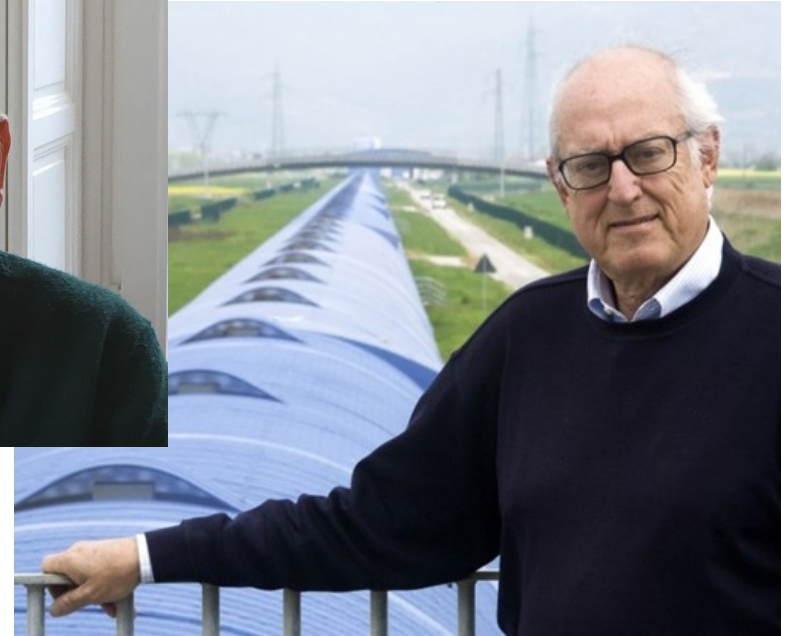
Premio Nobel dietro le quinte



Ronald Drever 1931-2017



Alain Brillet 1946



Adelberto Giazotto 1940-2017



La storia inizia qui

osservazione GW170817

PRL **119**, 161101 (2017) week ending
20 OCTOBER 2017

Selected for a Viewpoint in Physics
PHYSICAL REVIEW LETTERS

GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral

B. P. Abbott *et al.**

(LIGO Scientific Collaboration and Virgo Collaboration)

(Received 26 September 2017; revised manuscript received 2 October 2017; published 16 October 2017)

On August 17, 2017 at 12:41:04 UTC the Advanced LIGO and Advanced Virgo gravitational-wave detectors made their first observation of a binary neutron star inspiral. The signal, GW170817, was detected with a combined signal-to-noise ratio of 32.4 and a false-alarm-rate estimate of less than one per 8.0×10^4 years. We infer the component masses of the binary to be between 0.86 and $2.26 M_{\odot}$, in agreement with masses of known neutron stars. Restricting the component spins to the range inferred in binary neutron stars, we find the component masses to be in the range 1.17 – $1.60 M_{\odot}$, with the total mass of the system $2.74^{+0.04}_{-0.01} M_{\odot}$. The source was localized within a sky region of 28 deg^2 (90% probability) and had a luminosity distance of $40^{+8}_{-14} \text{ Mpc}$, the closest and most precisely localized gravitational-wave signal yet. The association with the γ -ray burst GRB 170817A, detected by Fermi-GBM 1.7 s after the coalescence, corroborates the hypothesis of a neutron star merger and provides the first direct evidence of a link between these mergers and short γ -ray bursts. Subsequent identification of transient counterparts across the electromagnetic spectrum in the same location further supports the interpretation of this event as a neutron star merger. This unprecedented joint gravitational and electromagnetic observation provides insight into astrophysics, dense matter, gravitation, and cosmology.

DOI: 10.1103/PhysRevLett.119.161101

Multi-messenger Observations of a Binary Neutron Star Merger

NSBH – PRIMA
OSSERVAZIONE
2020

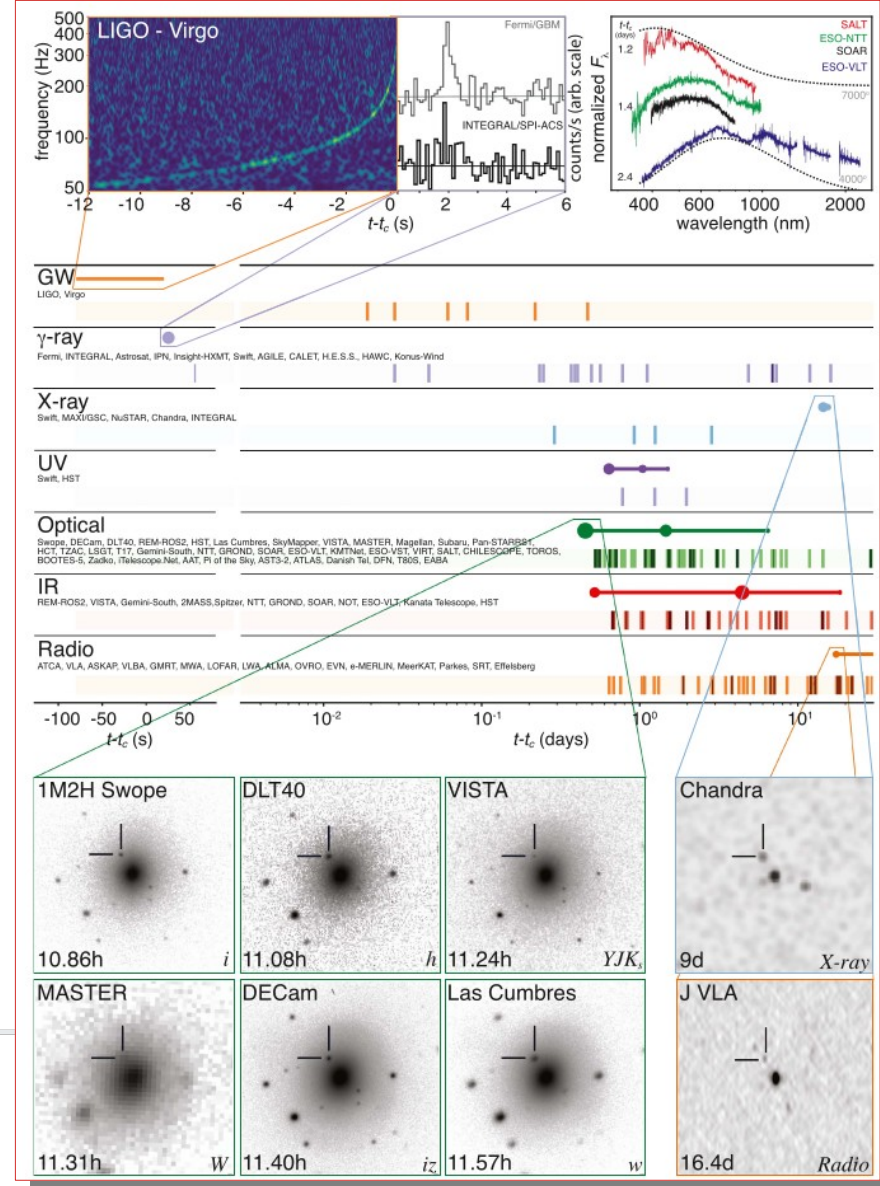
INIZIO DI O4
2023

2017

MULTI-MESSENGER
E BNS – PRIME
OSSERVAZIONI

2021

LIGO/Virgo/KAGRA
UPGRADE



Black Hole Neutron Star

pezzo mancante

Observation of Gravitational Waves from Two Neutron Star–Black Hole Coalescences

THE ASTROPHYSICAL JOURNAL LETTERS, 915:L5 (24pp), 2021 July 1

Abbott et al.

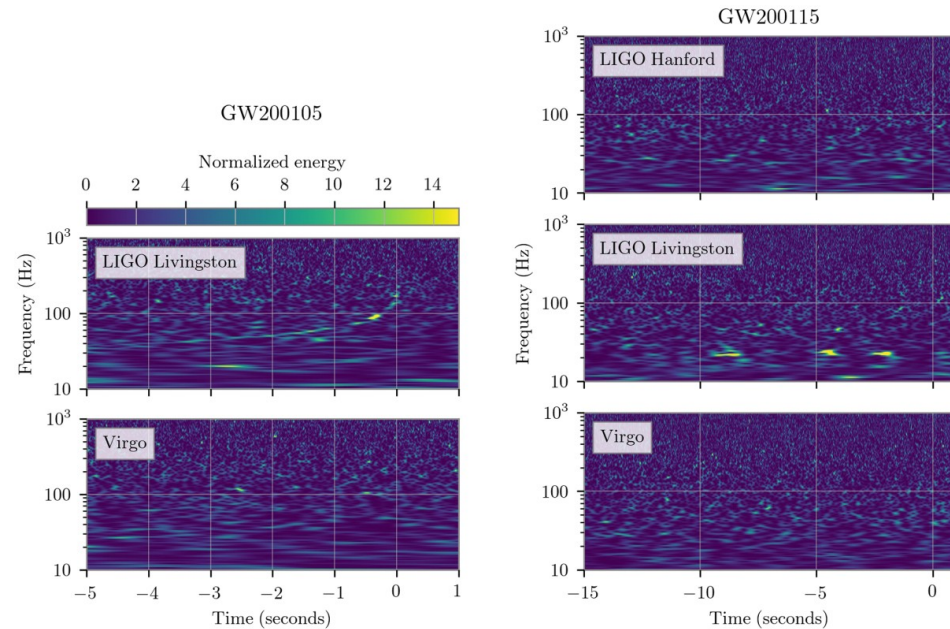


Figure 1. Time–frequency representations of the data containing GW200105 (left column) and GW200115 (right column). Times are shown relative to the signals’ merger times, 2020 January 5 at 16:24:26 UTC (left) and 2020 January 15 at 04:23:10 UTC (right). The amplitude scale of each time–frequency tile is normalized by the respective detector’s noise amplitude spectral density. The LIGO Livingston data of GW200105 show a track of excess power with increasing frequency. In the other panels, no similar tracks are visible, as the S/N in each of the detectors is lower and (for GW200115) the signal is longer. For GW200105, the LIGO Livingston data are shown after glitch subtraction. For GW200115, light-scattering noise is visible in LIGO Livingston below 25 Hz.

THE ASTROPHYSICAL JOURNAL LETTERS, 915:L5 (24pp), 2021 July 1

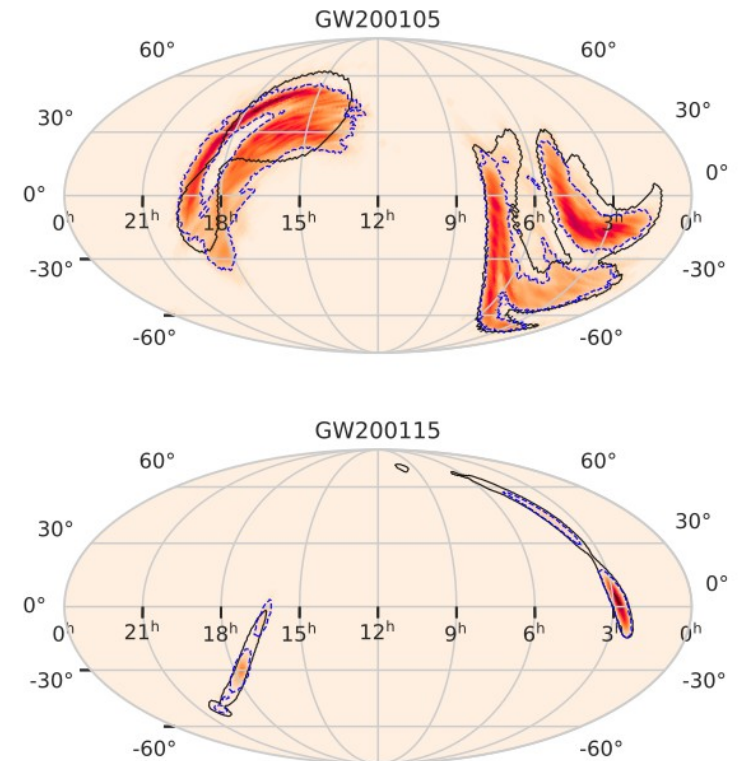
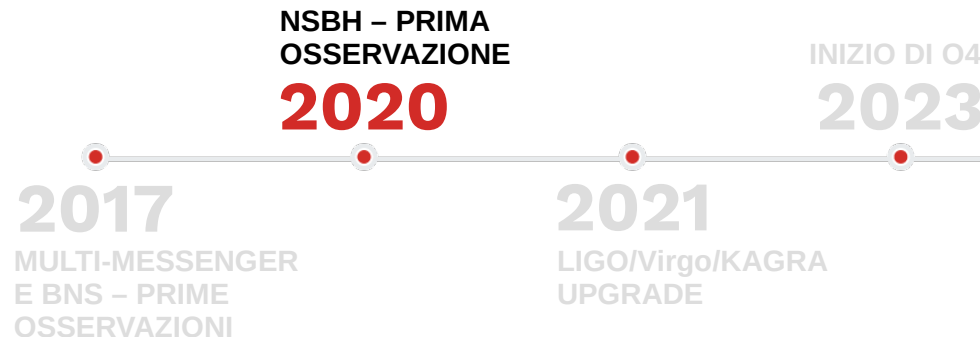


Figure 2. Sky localizations for GW200105 (top) and GW200115 (bottom) in terms of R.A. and decl. The thick solid contours show the 90% credible regions from the low-latency sky localization algorithm BAYESTAR (Singer & Price 2016). The shaded patch is the sky map obtained from the preferred high-spin analysis of Section 4, with the 90% credible regions bounded by the thin dotted contours.



Miglioramenti ai rivelatori

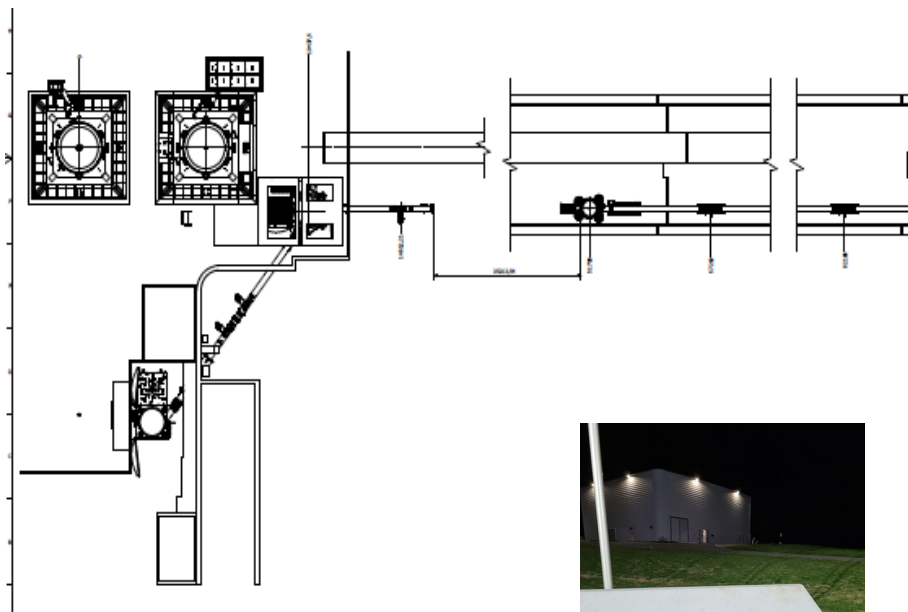
Virgo upgrade

- aggiunto Quantum Noise Reduction
- costruzione di una cavità ottica (FLT) di ~300m



Miglioramenti ai rivelatori

Virgo upgrade



NSBH – PRIMA
OSSERVAZIONE
2020

INIZIO DI O4
2023

2017

MULTI-MESSENGER
E BNS – PRIME
OSSERVAZIONI

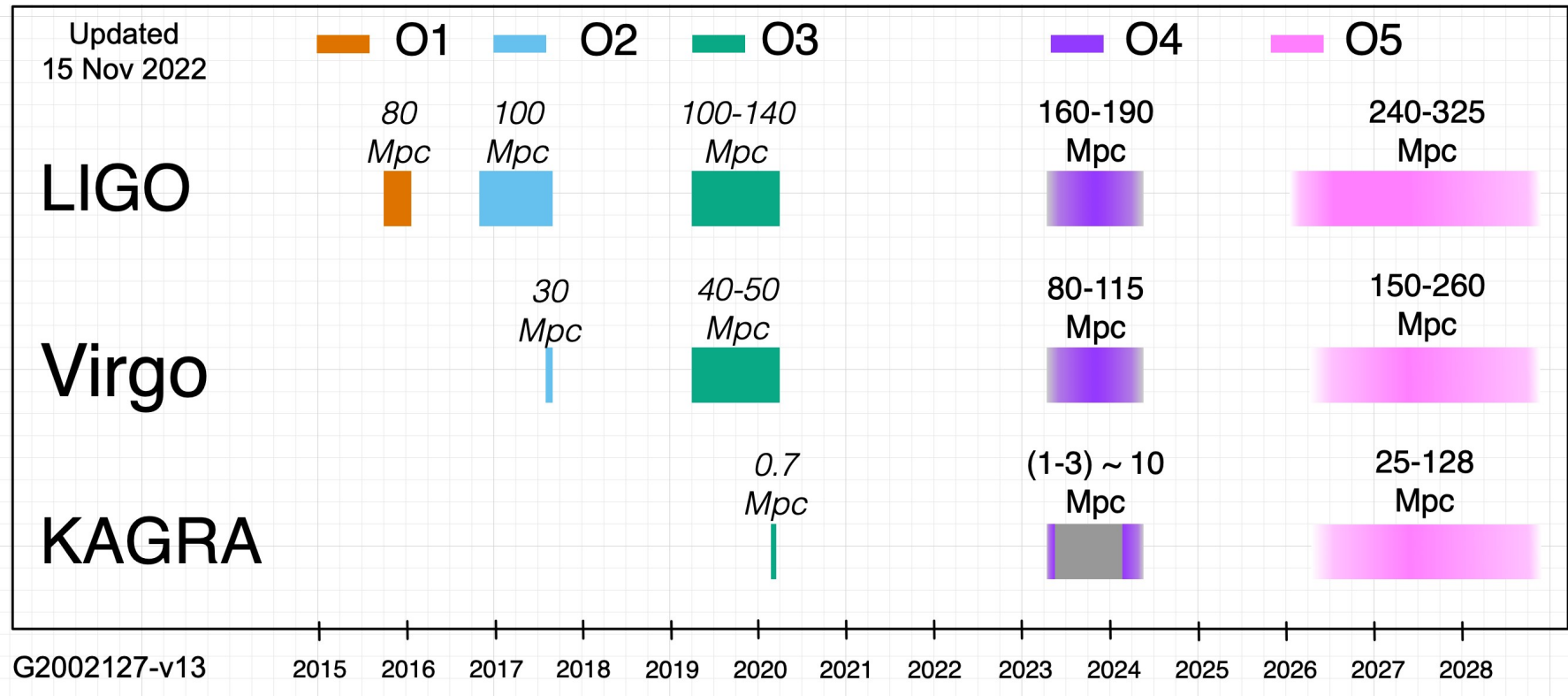
2021

LIGO/Virgo/KAGRA
UPGRADE

https://www.virgo-gw.eu/Laser/upgrade_1_F.html

O4

osservazioni



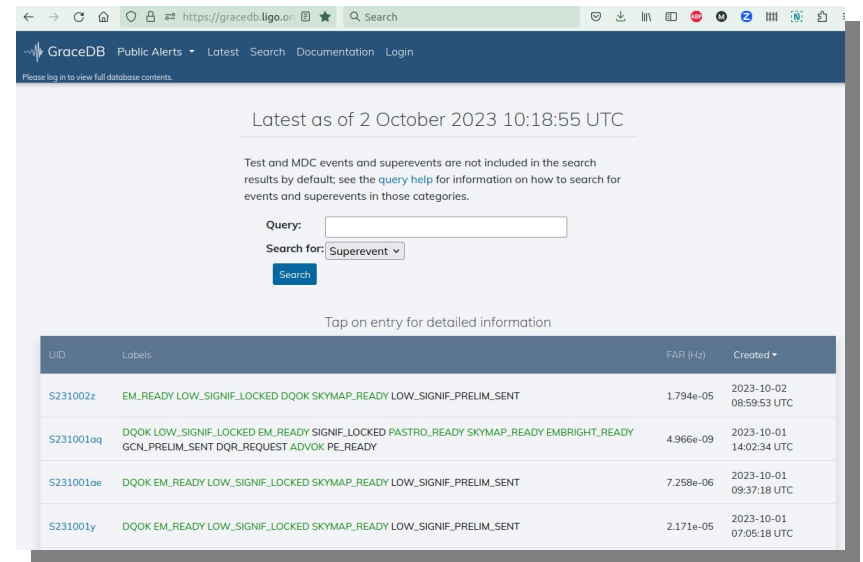
O4

osservazioni – annunci pubblici

Durante O4 le rivelazioni considerate importanti vengono pubblicate per permettere il follow-up elettromagnetico.

Risorse on-line:

- <https://gracedb.ligo.org/latest/>
- <https://virgo.pg.infn.it/maps/>

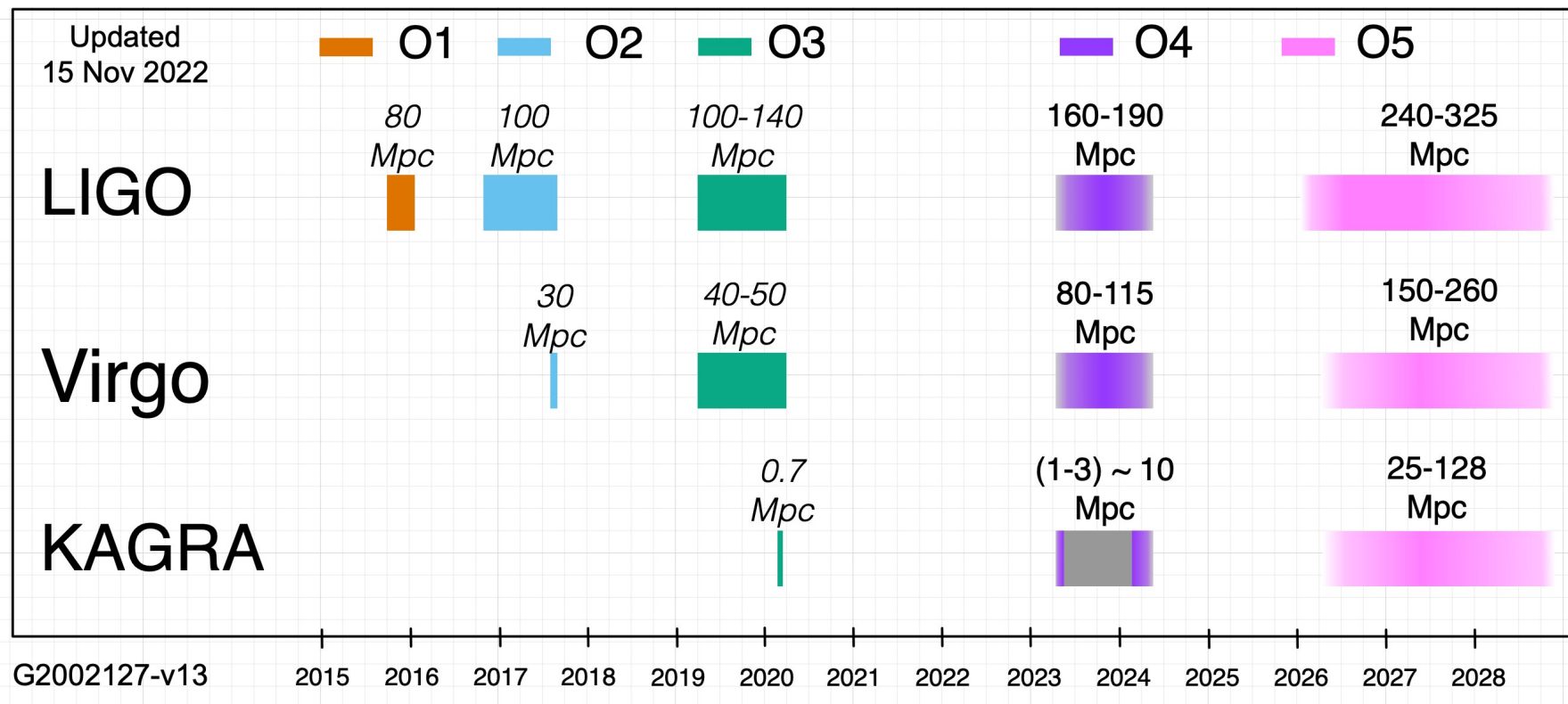


The screenshot shows the GraceDB Public Alerts interface. It includes a search bar, a 'Search for' dropdown set to 'Superevent', and a 'Search' button. Below the search area is a table of events with columns for UID, Labels, FAR (Hz), and Created. The table lists four events from October 2023.

UID	Labels	FAR (Hz)	Created
S231002z	EM_READY LOW_SIGNIF_LOCKED DQOK SKYMAP_READY LOW_SIGNIF_PRELIM_SENT	1.794e-05	2023-10-02 08:59:53 UTC
S231001oq	DQOK LOW_SIGNIF_LOCKED EM_READY SIGNIF_LOCKED PASTRO_READY SKYMAP_READY EMBRIGHT_READY GCN_PRELIM_SENT DQR_REQUEST ADVOK PE_READY	4.966e-09	2023-10-01 14:02:34 UTC
S231001oe	DQOK EM_READY LOW_SIGNIF_LOCKED SKYMAP_READY LOW_SIGNIF_PRELIM_SENT	7.258e-06	2023-10-01 09:37:18 UTC
S231001y	DQOK EM_READY LOW_SIGNIF_LOCKED SKYMAP_READY LOW_SIGNIF_PRELIM_SENT	2.171e-05	2023-10-01 07:05:18 UTC



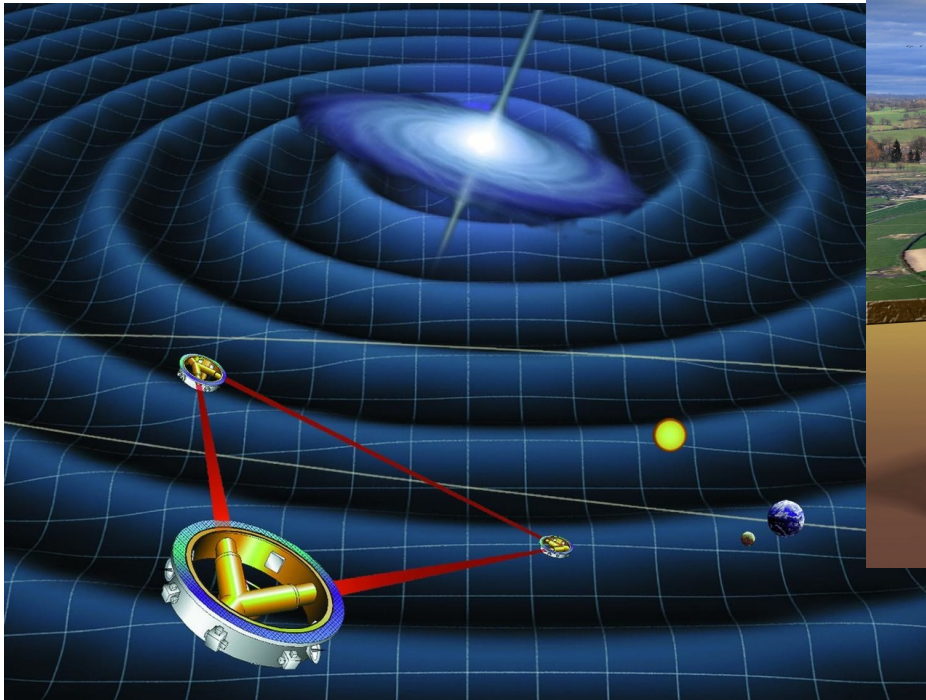
Periodi osservativi di LIGO, Virgo e KAGRA



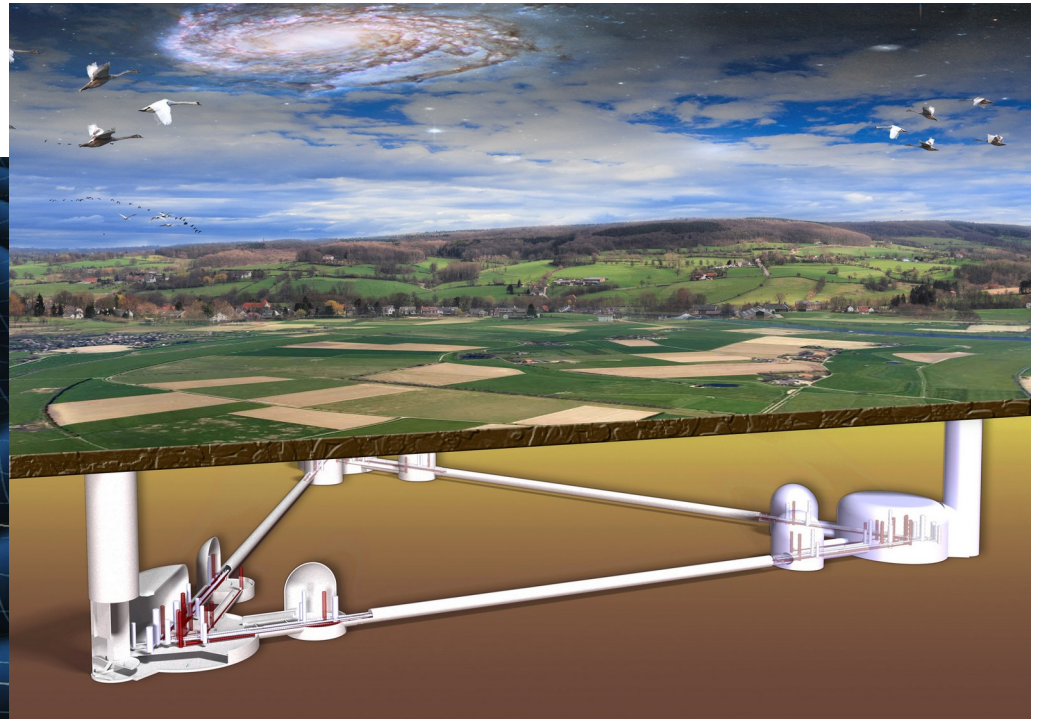
- GWTC-1: A Gravitational-Wave Transient Catalog of Compact Binary Mergers Observed by LIGO and Virgo during the First and Second Observing Runs
- GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo during the First Half of the Third Observing Run
- GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run

Futuro

EINSTEIN TELESCOPE



LISA



Referenze

- 1) <https://www.nobelprize.org/prizes/physics/2017/summary/>
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